

HOMEWORK III
NUMERICAL METHODS
FALL 2025

DUE TUESDAY, OCT 7

Instructions: Consider the following problems.

Prob 1. What linear function $y = ax + b$ approximates $f(x) = \sin x$ best in the vicinity of $x = \pi/4$? How does this problem relate to Newton's Method?

Prob 2. Locate the root of $f(x) = e^{-x} - \cos x$ that is nearest $\pi/4$.

Prob 3. Importance of choosing x_0 : for what starting values will Newton's Method converge for solving $f(x) = 0$? Here $f(x)$ is given by

$$f(x) = \frac{x^2}{1 + x^2}.$$

Computer Prob 1. Use Newton's method to approximate, to within 10^{-8} , the root of the equation $x^3 = x^2 + x + 1$.

Hint: You are allowed to guide your initial guess, x_0 , by looking at the graph of $f(x)$.

Computer Prob 2. Using the bisection method, find the positive root of

$$2x(1 + x^2)^{-1} = \arctan x.$$

Computer Prob 3. Solve the system of non-linear equations by first eliminating y and then solving the resulting equation in x by Newton's method. Initiate your Newton's method with $x_0 = 1.0$. The system is given by

$$\begin{cases} x^3 - 2xy + y^7 - 4x^3y = 5 \\ y \sin x + 3x^2y + \tan x = 4. \end{cases}$$

Note1: The programming should be done in terms of a Script and a function file in Matlab. Your submission should include a folder named "YourLastName" that contains 1) a .txt file with basic instructions/description of your script.m and your function.m file 2) script.m file and 3) function.m file. It is important to compress your folder before submitting it on Canvas. Try something like Winzip or Winrar. **Always include a Project.PDF file + your computer code files on your submission.** . Use graphical representation whenever possible to support summaries.

Note2: Show all your work. **Note3:** Yes, we will be running your codes.

*****Please follow instructions carefully - submit on time/no late acceptance *****
